

Exploratory Data Analysis

COMPARING GROUPS AND RELATIONSHIPS

Sergio I. García-Ríos

Fall 2026

PA 311N: Data Analysis for Public Policy

LBJ School of Public Affairs

Where we left off

Last time we focused on exploring **one variable at a time**.

Categorical variables

- Counts
- Proportions
- Bar plots

Quantitative variables

- Histograms
- Mean and median
- Spread and unusual values

Today we ask a harder question: **How do variables relate to one another?**

The basic EDA question

Suppose we observe a difference or a pattern.

Is it **real**, **important**, or just an artifact of how we looked at the data?

Exploratory data analysis helps us:

- compare groups;
- identify relationships;
- find unusual observations;
- discover alternative explanations;
- decide what to investigate next.

Start with the types of variables

The plot we choose depends on the variables we are comparing.

Variable 1	Variable 2	Useful display
Categorical	Categorical	Two-way table / segmented bar plot
Categorical	Quantitative	Side-by-side boxplots
Quantitative	Quantitative	Scatterplot

First identify the variable types. Then choose the tool.

Two categorical variables

Two categorical variables: compare proportions

Imagine a survey asking whether respondents support a housing policy.

	Support	Oppose	Total
Renters	72	28	100
Homeowners	54	46	100

Your Turn

Which group is more supportive? What quantity did you compare?

Counts are not always enough

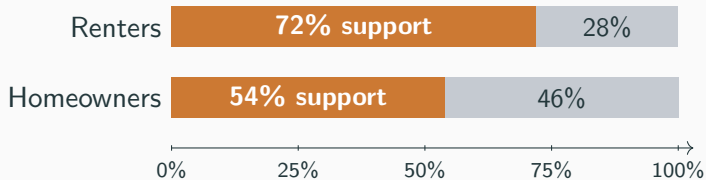
Now suppose the group sizes differ.

	Support	Oppose	Total
Renters	144	56	200
Homeowners	81	69	150

Comparing raw counts can mislead when group sizes differ.

Compare conditional proportions: support *within* each group.

A segmented bar plot makes the comparison visible



Segmented bars emphasize **composition within groups**.

Your Turn

What is the substantive difference between the two groups?

Ask the right percentage question

For a two-way table, there is more than one possible percentage.

- What percent of **renters** support the policy?
- What percent of **supporters** are renters?

These are **different questions** with different denominators.

Always ask: “**Percent of what?**”

**One categorical and one
quantitative variable**

Comparing a quantitative outcome across groups

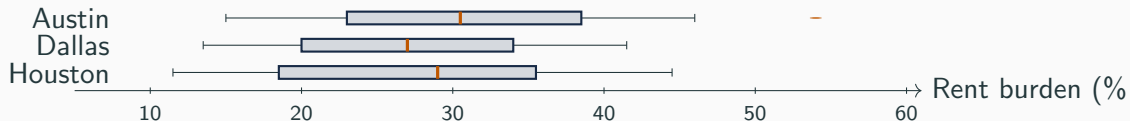
Suppose we want to compare rent burden across three metropolitan areas.

- Outcome: percent of household income spent on rent
- Group: metropolitan area

A single mean for each city can hide important differences in the distributions.

We want to compare center, spread, and unusual observations.

Side-by-side boxplots



Illustrative distributions.

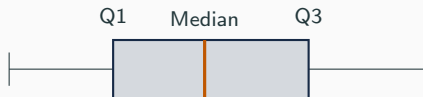
Your Turn

Which metro has the highest typical rent burden? Which shows the most spread?

Where is the unusual observation?

What a boxplot summarizes

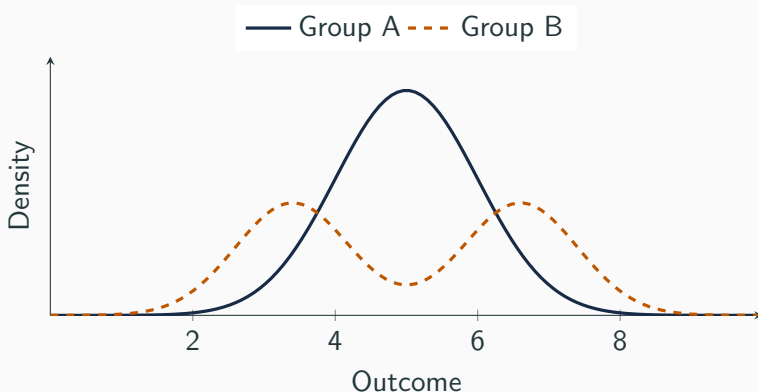
- Median
- First and third quartiles
- Interquartile range (IQR)
- Potential outliers



Boxplots are especially useful for comparing the **same outcome across several groups**.

Do not reduce every comparison to the mean

Two groups can have the same average but very different distributions.

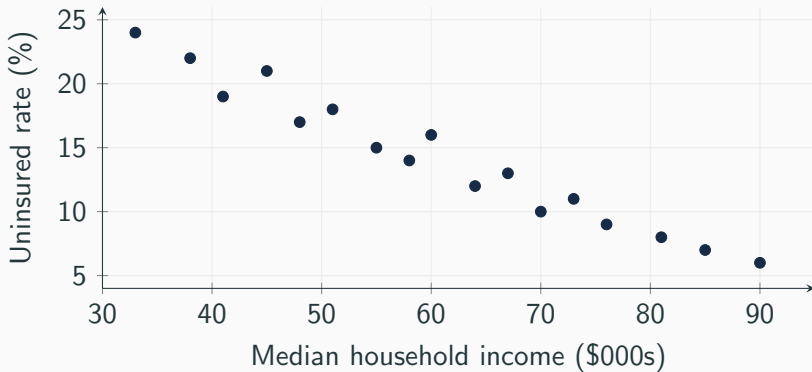


Same center does not mean same distribution.

Two quantitative variables

Two quantitative variables: use a scatterplot

Suppose we compare county median income with the uninsured rate.



Illustrative county-level data.

Describe a scatterplot before explaining it

When looking at two quantitative variables, describe:

1. **Direction**: positive, negative, or no clear direction
2. **Form**: roughly linear, curved, clustered
3. **Strength**: tight or diffuse pattern
4. **Unusual observations**: points that do not fit the pattern

Only after describing the pattern should we ask **why** it might exist.

Association is not explanation

Suppose higher-income counties tend to have lower uninsured rates.

That is an **association**.

It does **not** tell us by itself whether higher income causes lower uninsured rates.

Other differences may matter:

- age composition;
- employment structure;
- state policy;
- urbanization;
- access to public programs.

A third variable can change the story

A variable related to both the explanatory variable and the outcome can distort the relationship we see.

This is the idea of a **confounding variable**.

EDA should make us ask:

What happens when we break the data into meaningful groups?

Simpson's paradox

An apparent admissions gap

UC Berkeley graduate admissions, 1973:

	Admitted	Rejected	Admit rate
Men	1,198	1,493	44.5%
Women	557	1,278	30.4%

Your Turn

Looking only at this table, what conclusion might you draw? What information is missing?

Source: `datasets::UCBAdmissions`; Bickel, Hammel, and O'Connell (1975).

Now compare within departments

Dept.	Men		Women	
	Admitted	Rate	Admitted	Rate
A	512	62.1%	89	82.4%
B	353	63.0%	17	68.0%
C	120	36.9%	202	34.1%
D	138	33.1%	131	34.9%
E	53	27.7%	94	23.9%
F	22	5.9%	24	7.0%

The aggregate difference changes substantially once department is considered.

Why did the aggregate pattern appear?

Applicants were distributed differently across departments with very different admission rates.

- Women applied more often to departments with lower overall admission rates.
- Men applied more often to departments with higher overall admission rates.

The aggregate relationship was partly a product of composition.

This is a classic example of **Simpson's paradox**: a relationship can weaken, disappear, or reverse after conditioning on another variable.

The lesson is bigger than Simpson's paradox

Whenever you compare groups, ask:

- Are the groups composed differently?
- Is there an important third variable?
- Would the pattern change within subgroups?
- Are we comparing like with like?

Good EDA is skeptical of the first pattern you see.

Patterns and uncertainty

Observed differences are not automatically meaningful

Imagine 52% support in one sample and 48% in another.

Is the difference:

- a stable population difference?
- random sample-to-sample variation?
- a consequence of measurement or selection?

EDA tells us **what we observed**.

Inference will later help us evaluate **how much uncertainty surrounds it**.

Description, association, causation

Keep these three questions separate.

Question	Example
Description	What is the uninsured rate?
Association	Do uninsured rates differ by income?
Causation	Would changing a policy reduce the uninsured rate?

EDA is essential for all three, but **EDA alone does not establish causation.**

A practical workflow

A practical EDA workflow

1. Identify the substantive question.
2. Identify the unit of observation and variable types.
3. Examine each variable on its own.
4. Compare variables using an appropriate display.
5. Look for unusual observations and subgroup differences.
6. Consider alternative explanations.
7. Write a descriptive conclusion that matches the evidence.

From question to plot



Cat. + Cat. → segmented bar / two-way table

Cat. + Quant. → side-by-side boxplots

Quant. + Quant. → scatterplot

Your turn: choose the display

For each question, choose a useful visualization.

1. Does policy support differ by party identification?
2. Do rents differ across neighborhoods?
3. Is county income related to the uninsured rate?

Your turn: choose the display

For each question, choose a useful visualization.

1. Does policy support differ by party identification?
2. Do rents differ across neighborhoods?
3. Is county income related to the uninsured rate?

1. Segmented bar plot

2. Side-by-side boxplots

3. Scatterplot

What you should be able to do now

Given a substantive question, you should be able to:

- identify the relevant variables and their types;
- choose an appropriate visualization;
- describe patterns in terms of center, spread, direction, and unusual observations;
- compare groups using proportions rather than raw counts when appropriate;
- recognize that a third variable can change an apparent relationship;
- avoid making causal claims from descriptive patterns alone.

Explore first.

Then decide what the data can actually support.